

D284

Software Engineering

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A team of 12 at an insurance company, working in two-week sprints, spends three days drawing UML class and sequence diagrams and an architecture sketch before anyone writes code. Which framework activity is this?

- A. Deployment
- B. Modeling
- C. Construction
- D. Planning

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A state agency, which has run the same mainframe since 1998, must update payroll to match a new law. The rules are fully written, fixed, and well understood, and the deadline is set. Which process model fits best?

- A. Prototyping
- B. Agile, because it is always the better choice
- C. Waterfall
- D. Spiral

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A startup changes its app's requirements every week based on user feedback. Which approach fits best?

- A. An agile approach with short iterations
- B. Full up-front design, then freeze the spec
- C. Waterfall
- D. V-model

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

An online catalog serving a library system with 40 branches has this requirement: "The system shall return search results within 2 seconds for 95% of queries." How is it classified?

- A. Functional, because it is about search
- B. An acceptance test, not a requirement
- C. Non-functional (performance)
- D. Not a requirement, since requirements can never contain numbers

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Which of these is a functional requirement?

- A. The system shall be written in Java.
- B. The system shall let a librarian renew a patron's loan.
- C. The system shall be available 99.9% of the time.
- D. The system shall encrypt stored patron data.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

At a firm with three product lines, marketing insists a new dashboard ship by June with 20 reports, and engineering says only 8 reports are possible by then. The analyst meets both groups to rank the reports and agree on a scope. Which requirements task is this?

- A. Validation
- B. Negotiation
- C. Elicitation
- D. Specification

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A 4,000-line payroll module computes pay, formats paper checks, and emails HR reports. It also reads and changes global variables inside the benefits module. What is the main design problem?

- A. The module has too many lines of code
- B. Low cohesion only; shared globals are fine
- C. High cohesion and low coupling
- D. Low cohesion and high coupling

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Which change lowers coupling between two modules?

- A.** Merge both modules into one.
- B.** Add more shared global variables.
- C.** Rename the modules more clearly.
- D.** Pass needed data through parameters of a defined interface instead of shared global variables.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Five reviewers spend three hours checking a new billing module's code line by line against the approved design specification. What is this activity?

- A.** Verification
- B.** Validation
- C.** User acceptance testing
- D.** Debugging

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Customers use a beta release for two weeks and confirm it actually solves their scheduling problem. What is this activity?

- A.** Code inspection
- B.** Verification
- C.** Unit testing
- D.** Validation

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A release is tied to a trade show in April. The manager proposes skipping requirements and design reviews to save two weeks and “catch everything in testing.” What is the best response?

- A.** Keep the reviews; defects found in testing or production cost far more to fix than those found early.
- B.** Move the reviews to after release.
- C.** Agree; testing catches every defect.
- D.** Agree; a defect costs the same to fix at any stage.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A developer makes a one-line fix to the tax calculation in an accounting app used by 600 small businesses. What testing should follow?

- A.** Customer acceptance testing only
- B.** Unit test only the changed line.
- C.** Regression testing: rerun existing tests to confirm nothing else broke.
- D.** None; a one-line fix is too small to break anything.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A team ~~of 12 at an insurance company, working in two-week sprints,~~ spends three days drawing UML class and sequence diagrams and an architecture sketch before anyone writes code. Which framework activity is this?

Target word: framework activity **Decides:** Creating design models before code.

A	Inverted operation	Delivery and feedback come at the end, not before code.
B	Correct	Analysis and design models belong to modeling.
C	Adjacent category	Construction is coding and testing.
D	Confusion set	Planning covers estimates, schedules, and risks.

2. Answer C Concept 2

A state agency, ~~which has run the same mainframe since 1998,~~ must update payroll to match a new law. The rules are fully written, fixed, and well understood, and the deadline is set. Which process model fits best?

Target word: model for stable requirements **Decides:** Requirements are fixed and clear.

A	Adjacent category	Used when requirements are unclear.
B	Absolute language	No model is best in every case.
C	Correct	Linear phases work when requirements will not change.
D	Confusion set	Built for large, high-risk projects needing repeated risk analysis.

3. Answer A Concept 2 · transfer

A startup changes its app's requirements every week based on user feedback. Which approach fits best?

Target word: model for changing requirements **Decides:** Frequent change and feedback.

A	Correct	Built to absorb frequent change.
B	Adjacent category	Change would be costly under a frozen spec.
C	Inverted operation	Assumes requirements are frozen early.
D	Confusion set	A verification-heavy variant of waterfall.

4. Answer C Concept 3

An online catalog ~~serving a library system with 40 branches~~ has this requirement: "The system shall return search results within 2 seconds for 95% of queries." How is it classified?

Target word: requirement type **Decides:** It states how fast, not what.

A	Confusion set	The search function is assumed; the requirement adds a speed limit.
B	Adjacent category	It can drive a test, but it is still a requirement.
C	Correct	Sets a quality level for an existing function.
D	Absolute language	Measurable requirements are preferred.

5. Answer B Concept 3 · transfer

Which of these is a functional requirement?

Target word: pick the functional one **Decides:** Describes a system behavior.

A	Adjacent category	An implementation constraint.
B	Correct	States something the system does.
C	Inverted operation	Availability is non-functional.
D	Confusion set	Security quality, non-functional.

6. Answer B Concept 4

~~At a firm with three product lines,~~ marketing insists a new dashboard ship by June with 20 reports, and engineering says only 8 reports are possible by then. The analyst meets both groups to rank the reports and agree on a scope. Which requirements task is this?

Target word: name the task **Decides:** Resolving conflicting stakeholder demands.

A	Confusion set	Checks that the written spec is correct and complete.
B	Correct	Reconciles conflicts and sets priorities.
C	Adjacent category	Gathers requirements; does not settle conflicts.
D	Dropped element	Writes up requirements after they are agreed.

7. Answer D Concept 5

~~A 4,000-line~~ payroll module computes pay, formats paper checks, and emails HR reports. It also reads and changes global variables inside the benefits module. What is the main design problem?

Target word: diagnose cohesion and coupling **Decides:** Many unrelated jobs plus shared globals.

A	Camouflage	Size is not the core design flaw.
B	Dropped element	Shared globals tightly couple modules.
C	Inverted operation	That is the goal, not this module.
D	Correct	Mixed jobs lower cohesion; shared globals raise coupling.

8. Answer D Concept 5 · transfer

Which change lowers coupling between two modules?

Target word: action that reduces coupling **Decides:** Explicit, narrow interfaces.

A	Adjacent category	Hides the coupling but lowers cohesion.
B	Inverted operation	Increases coupling.
C	Dropped element	Names do not change dependencies.
D	Correct	Dependencies become few and explicit.

9. Answer A Concept 6

~~Five reviewers spend three hours~~ checking a new billing module's code line by line against the approved design specification. What is this activity?

Target word: verification or validation **Decides:** Checked against a specification.

A	Correct	Checks that the product matches its spec.
B	Confusion set	Checks against user needs, usually with users.
C	Adjacent category	Done by customers on working software.
D	Dropped element	Fixes known defects; a review finds them.

10. Answer D Concept 6 • transfer

Customers use a beta release for two weeks and confirm it actually solves their scheduling problem. What is this activity?

Target word: verification or validation **Decides:** Checked against real user needs.

A	Confusion set	A review against standards and specs.
B	Inverted operation	Checks against the spec, not user needs.
C	Adjacent category	Tests individual components.
D	Correct	Confirms it is the right product.

11. Answer A Concept 7

~~A release is tied to a trade show in April.~~ The manager proposes skipping requirements and design reviews to save two weeks and “catch everything in testing.” What is the best response?

Target word: cost of late defects **Decides:** Defects found later cost many times more.

A	Correct	Early reviews reduce total cost and delay.
B	Adjacent category	Defects would reach customers first.
C	Absolute language	Testing never finds all defects.
D	Inverted operation	Cost rises sharply later.

12. Answer C Concept 8

~~A developer makes a one-line fix to the tax calculation in an accounting app used by 600 small businesses.~~ What testing should follow?

Target word: testing after a change **Decides:** Any change can break existing behavior.

A	Adjacent category	Checks fit for users, not side effects.
B	Dropped element	Misses effects on other features.
C	Correct	Catches side effects of the change.
D	Camouflage	Small changes can cause large regressions.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	11	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Generic process framework	Five framework activities: communication, planning, modeling, construction (code and test), and deployment.	Calling design diagrams “construction” or “planning.”	A described activity and “which framework activity is this?”
2	Choosing a process model	Waterfall fits stable, well-understood requirements; agile fits requirements that change often and need fast feedback.	Believing one model is always best.	How stable and clear the requirements are.
3	Functional vs. non-functional requirements	Functional requirements say what the system does; non-functional requirements say how well (performance, security, availability, usability).	Calling a performance target functional because it mentions a feature.	“Within 2 seconds,” “99.9% uptime,” “encrypted.”
4	Requirements engineering tasks	Inception, elicitation, elaboration, negotiation, specification, validation, and management; negotiation resolves conflicts and sets priorities.	Treating conflict resolution as more elicitation.	Stakeholders want incompatible things.
5	Cohesion and coupling	Good modules have high cohesion (one focused job) and low coupling (few, explicit dependencies).	Reversing which should be high and which low.	A module doing many jobs or sharing global data.
6	Verification vs. validation	Verification: are we building the product right (meets the spec)? Validation: are we building the right product (meets user needs)?	Swapping the two.	Checking against a document vs. checking with real users.
7	Cost of late defects	Defects cost far more to fix the later they are found, so early reviews pay for themselves.	Skipping reviews to save time and relying on testing.	A plan to cut early quality steps.
8	Regression testing	After any change, rerun existing tests to confirm nothing that used to work is now broken.	Testing only the changed line.	A fix or change to working software.

Course scope (public description): Software Engineering introduces the concepts of software engineering to students who have completed the core courses in programming and project management. The principles build on previously acquired concepts, switching the emphasis from programming simple routines to engineering robust and scalable software solutions. This course does not cover programming, but it provides an overview of software engineering processes and their challenging nature, focusing on the need for a disciplined approach to software engineering. A generic process framework provides the groundwork for formal process models. Prescriptive process models such as the Waterfall Model and Agile Development are included. This course also introduces the elements and phases of software engineering, including requirements engineering, design concepts, and software quality. There are no prerequisites for this course.